Volume Iii Integration Reference

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Integration

0.1 Partitions and Tags

The Geometric Problem

Definition 0.1.1 (Partition of a Compact Interval)**.** A *partition* of $[a, b]$ is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned}\operatorname{Partition}(P, a, b) \iff {} & \exists n \in \mathbb{N}\, \exists x_0, \dots, x_n \in \mathbb{R} \\ & \big(P = \{x_0, \dots, x_n\} \wedge x_0 = a \wedge x_n = b\big) \\ & \wedge \forall i \, \big(1 \le i \le n \Rightarrow x_{i-1} < x_i\big).\end{aligned}$$

Definition 0.1.2 (Subinterval Width)**.** The i-th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

Definition 0.1.3 (Mesh of a Partition)**.** The *mesh* or *norm* of P is the width of its fattest slab:

$$\|P\| = \max_{1 \le i \le n} \Delta x_i.$$

Definition 0.1.4 (Tagged Partition)**.** A *tagged partition* is a partition $P = \{x_0, \dots, x_n\}$ together with points $\xi_i \in [x_{i-1}, x_i]$. The tag is the point where the height $f(\xi_i)$ is sampled.

Definition 0.1.5 (Refinement of a Partition)**.** A partition P' *refines* P when $P \subseteq P'$. Refinement adds cut-points; it does not move or delete existing cut-points.

Lemma 0.1.6 (Common Refinement of Two Partitions)**.** *Any two partitions P and P' of $[a, b]$ have a common refinement: order the finite set $P \cup P'$ increasingly.*

0.2 The Cauchy Integral

Geometric Intuition

Definition 0.2.1 (Left-Endpoint Cauchy Sum)**.** For $f : [a, b] \to \mathbb{R}$ and $P = \{a = x_0 < \cdots < x_n = b\}$, define

$$C(f, P) = \sum_{i=1}^{n} f(x_{i-1})\, \Delta x_i.$$

Definition 0.2.2 (Cauchy Integral)**.** The function f is *Cauchy-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall P\ \big(\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon\big).$$

The value L is denoted $\int_a^b f(x)\, dx$.

Proposition 0.2.3 (Cauchy Integral of a Constant)**.** *If $f(x) = c$ on $[a, b]$, then $\int_a^b c\,dx = c(b - a)$.*

Theorem 0.2.4 (Linearity of the Cauchy Integral)**.** *If f and g are Cauchy-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem 0.2.5 (Monotonicity of the Cauchy Integral)**.** *If f and g are Cauchy-integrable on $[a, b]$ and $f(x) \leq g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \leq \int_a^b g\,dx.$$

Corollary 0.2.6 (Bounds for the Cauchy Integral)**.** *If f is continuous on $[a, b]$, $m = \min_{[a,b]} f$, and $M = \max_{[a,b]} f$, then*

$$m(b - a) \leq \int_a^b f\,dx \leq M(b - a).$$

Theorem 0.2.7 (Triangle Inequality for the Cauchy Integral)**.** *If f and $|f|$ are Cauchy-integrable on $[a, b]$, then*

$$\left| \int_a^b f\,dx \right| \leq \int_a^b |f|\,dx.$$

Theorem 0.2.8 (Interval Additivity for the Cauchy Integral)**.** *If $c \in [a, b]$ and f is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Definition 0.2.9 (Oscillation on an Interval)**.** For bounded f on an interval I, define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if $\omega_f(I)$ is small, every sample height on I is close to every other sample height.

Theorem 0.2.10 (Continuous Functions Are Cauchy-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Cauchy-integrable.*

Theorem 0.2.11 (Tag Independence for Continuous Functions)**.** *If f is continuous on $[a, b]$, then every choice of tags gives the same limit as the left-endpoint sums as $\|P\| \to 0$.*

0.3 The Riemann Integral

Geometric Intuition

Definition 0.3.1 (Riemann Sum)**.** For a tagged partition (P, ξ), define

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Definition 0.3.2 (Riemann Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Riemann-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall (P, \xi)\ \big(\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon\big).$$

Theorem 0.3.3 (Continuous Functions Are Riemann-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.*

Theorem 0.3.4 (Linearity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem 0.3.5 (Monotonicity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \le \int_a^b g\,dx.$$

Theorem 0.3.6 (Triangle Inequality for the Riemann Integral)**.** *If f is Riemann-integrable on $[a, b]$, then $|f|$ is Riemann-integrable and*

$$\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx.$$

Theorem 0.3.7 (Interval Additivity for the Riemann Integral)**.** *If $c \in [a, b]$, then f is Riemann-integrable on $[a, b]$ if and only if it is Riemann-integrable on $[a, c]$ and $[c, b]$, and in that case*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Theorem 0.3.8 (Cauchy Criterion for Riemann Integrability)**.** *A bounded function is Riemann-integrable if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that any two tagged partitions (P, ξ) and (Q, η) with $\|P\|, \|Q\| < \delta$ have*

$$|S(f, P, \xi) - S(f, Q, \eta)| < \varepsilon.$$

0.4 The Darboux Integral

Geometric Intuition

Definition 0.4.1 (Lower and Upper Darboux Sums)**.** For bounded f on $[a, b]$, set

$$m_i = \inf_{[x_{i-1}, x_i]} f, \qquad M_i = \sup_{[x_{i-1}, x_i]} f,$$

and define

$$L(f, P) = \sum_i m_i \Delta x_i, \qquad U(f, P) = \sum_i M_i \Delta x_i.$$

Lemma 0.4.2 (Darboux Sums Under Refinement)**.** *If P' refines P, then*

$$L(f, P) \le L(f, P') \le U(f, P') \le U(f, P).$$

Definition 0.4.3 (Darboux Integrability)**.** The lower and upper integrals are

$$\underline{I} = \sup_P L(f, P), \qquad \overline{I} = \inf_P U(f, P).$$

The function f is *Darboux-integrable* if $\underline{I} = \overline{I}$.

Theorem 0.4.4 (Darboux Criterion)**.** *A bounded function f is Darboux-integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$U(f, P) - L(f, P) < \varepsilon.$$

Theorem 0.4.5 (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions on $[a, b]$, Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

Theorem 0.4.6 (Continuous Functions Are Darboux-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem 0.4.7 (Monotone Functions Are Darboux-Integrable)**.** *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem 0.4.8 (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable)**.** *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Theorem 0.4.9 (Sums and Scalar Multiples of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then $\alpha f + \beta g$ is Darboux-integrable for all $\alpha, \beta \in \mathbb{R}$.*

Theorem 0.4.10 (Products of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then fg is Darboux-integrable on $[a, b]$.*

Theorem 0.4.11 (Absolute Values of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, then $|f|$ is Darboux-integrable on $[a, b]$.*

Theorem 0.4.12 (Continuous Images of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, φ is continuous on an interval containing $f([a, b])$, and f is bounded, then $\varphi \circ f$ is Darboux-integrable on $[a, b]$.*

0.5 The Riemann–Stieltjes Integral

Geometric Intuition

Definition 0.5.1 (Total Variation)**.** The total variation of α on $[a, b]$ is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

Definition 0.5.2 (Bounded Variation)**.** The function α has *bounded variation* on $[a, b]$ if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.

Proposition 0.5.3 (Monotone Functions Have Bounded Variation)**.** *If α is monotone on $[a, b]$, then α has bounded variation.*

Definition 0.5.4 (Riemann–Stieltjes Integral)**.** The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big).$$

When these sums have a tag-independent limit as $\|P\| \to 0$, the limit is denoted $\int_a^b f\, d\alpha$.

Theorem 0.5.5 (Continuous Integrand and BV Integrator)**.** *If f is continuous on $[a, b]$ and α has bounded variation, then $\int_a^b f\, d\alpha$ exists.*

Theorem 0.5.6 (Bilinearity of the Riemann–Stieltjes Integral)**.** *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g)\, d\alpha = \lambda \int_a^b f\, d\alpha + \mu \int_a^b g\, d\alpha$$

and

$$\int_a^b f\, d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f\, d\alpha + \mu \int_a^b f\, d\beta.$$

Theorem 0.5.7 (Interval Additivity for the Riemann–Stieltjes Integral)**.** *If $c \in [a, b]$ and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f\, d\alpha = \int_a^c f\, d\alpha + \int_c^b f\, d\alpha.$$

Theorem 0.5.8 (Riemann–Stieltjes Integration by Parts)**.** *If f and α are such that one of $\int_a^b f\, d\alpha$ and $\int_a^b \alpha\, df$ exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f\, d\alpha + \int_a^b \alpha\, df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem 0.5.9 (Reduction to Riemann Integration for a C^1 Integrator)**.** *If $\alpha \in C^1([a, b])$ and f is Riemann-integrable on $[a, b]$, then*

$$\int_a^b f\, d\alpha = \int_a^b f(x)\alpha'(x)\, dx.$$

Theorem 0.5.10 (Step Integrators Give Finite Sums)**.** *If α is constant except for finitely many jumps at points c_k and f is continuous at each c_k, then*

$$\int_a^b f\, d\alpha = \sum_k f(c_k)\, \Delta\alpha(c_k).$$

0.6 The Henstock–Kurzweil Integral

Where This Sits

Geometric Intuition

Definition 0.6.1 (Gauge on an Interval)**.** A *gauge* on $[a, b]$ is a function

$$\delta : [a, b] \to (0, \infty).$$

Definition 0.6.2 (δ-Fine Tagged Partition)**.** A tagged partition (P, ξ) is *δ-fine* if

$$[x_{i-1}, x_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Definition 0.6.3 (Henstock–Kurzweil Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Henstock–Kurzweil integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a, b] \to (0, \infty)\ \forall (P, \xi)\ \big((P, \xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big).$$

The value is denoted (HK) $\int_a^b f$.

Lemma 0.6.4 (Cousin's Lemma)**.** *For every gauge δ on $[a, b]$, there exists a δ-fine tagged partition of $[a, b]$.*

Theorem 0.6.5 (Riemann Integrable Implies Henstock–Kurzweil Integrable)**.** *If f is Riemann-integrable on $[a, b]$, then f is Henstock–Kurzweil integrable on $[a, b]$, and the two integrals have the same value.*

Lemma 0.6.6 (Straddle Lemma)**.** *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then for every $\varepsilon > 0$ there is $\delta(\xi) > 0$ such that whenever $u \le \xi \le v$, $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$,*

$$|F(v) - F(u) - f(\xi)(v - u)| \le \varepsilon(v - u).$$

Theorem 0.6.7 (Fundamental Theorem for the Henstock–Kurzweil Integral)**.** *If $F : [a, b] \to \mathbb{R}$ is differentiable at every point and $F' = f$, then f is Henstock–Kurzweil integrable and*

$$(\text{HK}) \int_a^b f = F(b) - F(a).$$

Theorem 0.6.8 (Continuous Functions Are Henstock–Kurzweil Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Henstock–Kurzweil integrable.*

0.7 The McShane Integral

Where This Sits

Geometric Intuition

Definition 0.7.1 (McShane-Fine Tagged Partition)**.** Let $\delta : [a,b] \to (0,\infty)$ be a gauge. A finite tagged partition $\{([u_i,v_i],\xi_i)\}_{i=1}^n$ is *McShane δ-fine* if the intervals $[u_i,v_i]$ are non-overlapping, cover $[a,b]$, each $\xi_i \in [a,b]$, and

$$[u_i,v_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Unlike an HK tagged partition, the tag ξ_i is not required to lie in $[u_i,v_i]$.

Definition 0.7.2 (McShane Integral)**.** A function $f : [a,b] \to \mathbb{R}$ is *McShane integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a,b] \to (0,\infty)\ \forall (P,\xi)\ \big((P,\xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f,P,\xi) - A| < \varepsilon\big),$$

where

$$S(f,P,\xi) = \sum_i f(\xi_i)(v_i - u_i).$$

Theorem 0.7.3 (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a,b] \subseteq \mathcal{M}[a,b] \subseteq \mathcal{HK}[a,b],$$

and the integral values agree whenever a function belongs to the smaller class.

Theorem 0.7.4 (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b],$$

and the McShane and Lebesgue integral values agree.

0.8 Measure Zero and the Lebesgue Criterion

Geometric Intuition

Definition 0.8.1 (Measure Zero)**.** A set $E \subseteq \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are open intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \qquad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

Definition 0.8.2 (Point Oscillation)**.** For bounded f on $[a,b]$, define

$$\omega_f(x) = \inf_{\delta>0} \omega_f\big((x-\delta, x+\delta) \cap [a,b]\big).$$

Then f is continuous at x exactly when $\omega_f(x) = 0$.

Theorem 0.8.3 (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a,b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a,b] : \omega_f(x) > 0\}$$

has measure zero.

Proofs